

Week 5 Office Hour Notes

5.0 Errata

9/30/24 12:00 PM. Made sure everything was correctly formatted with bolded notation and made a few minor clarifications.

10/4/24 12:00 PM. M

5.1 Boolean Algebra

Definition 5.1.1. A **Boolean algebra** is a set A equipped with two binary operations $\wedge$ (called "meet") $\vee$ (called "join"), a unary operation $\neg$ (called "complement"), and two elements 0 and 1 (called "bottom" and "top"), such that the following axioms hold

$$a \vee (b \vee c) = (a \vee b) \vee c, a \wedge (b \wedge c) = (a \wedge b) \wedge c \text{ (associativity)}$$

$$a \vee b = b \vee a, a \wedge b = b \wedge a \text{ (commutativity)}$$

$$a \vee (a \wedge b) = a, a \wedge (a \vee b) = a \text{ (adsorption)}$$

$$a \vee 0 = a, a \wedge 1 = a \text{ (identity)}$$

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c), a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \text{ (distributivity)}$$

$$a \vee \neg a = 1, a \wedge \neg a = 0 \text{ (complement)}$$

In Boolean algebras, we also have a [partially ordered set](#) (i.e. poset) defined on A by

$$a \leq b \iff a = b \wedge a \iff a \vee b = b$$

$$a < b \iff a \leq b \text{ but } a \neq b$$

We can think of $\leq$ as analogous to tautological implication $\models$.

%EXPLAIN THAT FURTHER

Examples 5.1.2.

1. Given a universal set U the set

$$\mathcal{P}(U) = \{A : A \subset U\},$$

is a Boolean algebra with respect to the $\wedge, \vee, \neg, 0, 1$ being the set operations

$\cap, \cup, (-)^c, \emptyset, U$, where

$$A^c := U \setminus A = \{x \in U : x \notin A\}.$$

%CHANGE THIS EXAMPLE TO OTHER ONES THAT MAKE SENSE; DO BOOLEAN ALGEBRA GENERATED BY SOMETHING, EQUIVALENCE CLASSES OF SENTENTIAL LOGIC, AND NATURAL NUMBER EXAMPLE

2. Given the set $[-1, 1]$, we have a Boolean algebra with respect to $\wedge, \vee, \neg, 0, 1$ defined by

$$a \wedge b := \min(a, b)$$

$$a \vee b := \max(a, b)$$

$$\neg a := -a$$

$$0 := -1$$

$$1 := 1$$

3. Given the set $\{0, 1\}$, we have a Boolean algebra with respect to $\wedge, \vee, \neg, 0, 1$ defined by

$$a \wedge b := \min(a, b) = \begin{cases} 1 & \text{if } a, b = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$a \vee b := \max(a, b) = \begin{cases} 1 & \text{if } a = 1 \text{ or } b = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\neg a := 1 - a = \begin{cases} 0 & \text{if } a = 1 \\ 1 & \text{if } a = 0 \end{cases}$$

$$0 := 0$$

$$1 := 1$$

%CALL THIS THE TRIVIAL BOOLEAN ALGEBRA

4. Given the set $\{T, F\}$, we have a Boolean algebra with respect to $\wedge, \vee, \neg, 0, 1$ defined by

$$a \wedge b := \begin{cases} T & \text{if } a, b = T \\ F & \text{otherwise} \end{cases}$$

$$a \vee b := \begin{cases} T & \text{if } a = T \text{ or } b = T \\ F & \text{otherwise} \end{cases}$$

$$\neg a := \begin{cases} F & \text{if } a = T \\ T & \text{if } a = F \end{cases}$$

$$0 := 0$$

$$1 := 1$$

Note that **parts 3.** and **4.** are really the same example, i.e., we can define a bijective function $1 \mapsto T, 0 \mapsto F$ that preserves all the relevant properties (which we call **Boolean Algebra Isomorphic**)

4.2 Boolean Functions and More on Truth Tables

Definition 5.2.1. A **Boolean Function** is a function $B : \{T, F\}^n \rightarrow \{T, F\}$. A Boolean Function $B_\varphi : \{T, F\}^n \rightarrow \{T, F\}$ with respect to a wff φ is the analogous function mapping a number of truth values equal to the amount of sentence symbols that φ contains to its truth table value (as discussed in **Section 1.2**, i.e. the [Week 2 Office Hour Notes](#)).

NOTE: There is an equivalent way of defining it recursively that I'll show in a later edit of these notes.

Examples 5.2.2. In a later edition I'll go over some examples.

Note that in the book, we define an inequality $<$ on $\{T, F\}$, by the single property that $F < T$ (i.e., we're ranking T higher than F) and we find that this inequality is exactly what $<$ ends up being with respect to the Boolean algebra $\{T, F\}$ as defined in **part 4** of **Examples 4.1.2**, and we can conceptualize **part 4** as we would **part 3**, with $<$ as it would be usually defined on $\{0, 1\} \subset \mathbb{R}$ (i.e. we also have the ranking $0 < 1$ in this inequality).

%TALK ABOUT BOOLEAN ALGEBRA HOMOMORPHISMS AND EXPLAIN HOW TRUTH ASSIGNMENTS ARE BOOLEAN ALGEBRA HOMOMORPHISMS

5.3 Disjunctive Normal Form and Boolean Completeness

Definition 5.3.1. A wff φ is in **Disjunctive Normal Form (DNF)** if φ , which utilizes sentence symbols $\{B_1, \dots, B_n\}$ is of the form

$$\alpha_1 \vee \dots \vee \alpha_N, \quad (5.1)$$

where α_j for each $1 \leq j \leq N$ is a wff of the form

$$\epsilon_{j,1} \wedge \dots \wedge \epsilon_{j,n} \quad (5.2)$$

where $\epsilon_{j,k}$ is either B_k or $\neg B_k$.

The **Disjunctive Normal Form (DNF)** of a wff φ (if it exists) is a wff φ_0 that is in disjunctive normal form that is tautologically equivalent, i.e., we have $\varphi \models \varphi_0$

Examples 5.3.2. In a later edition I'll go over some examples.

%GO OVER THESE EXAMPLES

Definition 5.3.2. A set of connectives C in sentential logic is **Boolean complete** (referred to Boolean complete in the notes) if every Boolean function B can be created by a Boolean function with respect to a wff φ using only sentence symbols plus the connectives in C , i.e., we have $B = B_\varphi$ for some such wff φ .

Note that disjunctive normal form is an intuitive way to talk about the truth values in a Boolean function. The advantages of the disjunctive normal form is that it directly communicates what the Boolean function explicitly is, as in, if φ is of the form (5.1), then we can determine B_φ simply by looking at the collection $\varphi_1, \dots, \varphi_N$. All $(X_1, \dots, X_n) \in \{T, F\}^n$ such that there exists some α_j of the form of (5.2), for some $1 \leq j \leq N$, such that $X_k = T$ iff $\epsilon_{j,k}$ is B_k (and not the negation) and $X_k = F$ otherwise, for each $1 \leq k \leq n$ yields $B_\varphi(X_1, \dots, X_n) = T$, and otherwise we have $B_\varphi(X_1, \dots, X_n) = F$.

For example, let's say we have φ as the wff $(A \wedge B) \vee (\neg A \wedge \neg B)$. Since a boolean function in this setting maps $\{T, F\} \times \{T, F\} \rightarrow \{T, F\}$, the disjunctive normal form tells us the truth/false value combinations that are true in the formula, i.e., if either A is true, and B is true or A is false and B is false, then φ is true, and otherwise any other combination is false. So the Boolean function B_φ is defined as follows:

$$B_\varphi : (T, T), (F, F) \mapsto T, (T, F), (F, T) \mapsto F$$

%GIVE PREAMBLE FOR THEOREM.

Theorem 5.3.3.

1. $\{\neg, \wedge, \vee\}$ is Boolean complete.
2. Every formula is tautologically equivalent to a formula in DNF

Outline of Proof.

1. Any Boolean Function can be created from the disjunctive normal form.

%ELABORATE ON THAT ARGUMENT

2. Follows immediately from **part 1.** and the idea that two formulas are tautologically equivalent iff their Boolean functions are equivalent.

Corollary 5.3.4. The sets $\{\neg, \wedge\}$, $\{\neg, \vee\}$ are Boolean complete.

Outline of Proof. Utilizing **Theorem 4.3.3**, along with **DeMorgan's Laws** to see that the DNF of a formula can be simplified further in terms of either the $\neg, \wedge$ or $\neg, \vee$ connectives. $\square$

5.4 Other Connectives in Sentential Logic

In Section 1.5 of Enderton, we talked about other possible connectives in sentential logic.

For example, we discussed the ternary connective $\#$, which represents the "majority operator" which spits out truth iff the majority of the input values are true, so the boolean function for $\#(A, B, C)$

A	B	C	$\#(A, B, C)$
F	F	F	F
F	F	T	F
F	T	F	F
F	T	T	T
T	F	F	F
T	F	T	T
T	T	F	T
T	T	T	T

We also have what's called 0-ary connectives, where it's represented by a Boolean function without any inputs (i.e. a single input), aka a constant, which one can think of a function $\{*\} \rightarrow \{T, F\}$, and there are obviously only two possibilities, i.e. the connectives defined by $\top, \perp$, where

$$\top(*) := T, \quad \perp(*) := F$$

(note that we can define $\top$ as $A \vee \neg A$ and $\perp$ as $A \wedge \neg A$)

We defined the $\downarrow$ connective to be the "neither nor" binary connective, i.e.

$$(\alpha \downarrow \beta) := \neg(\alpha \vee \beta)$$

By **Theorem 5.3.3 part 1**, we found that $\{\neg, \vee, \wedge\}$ is complete, so all the connective can be talked about in terms of $\neg, \vee, \wedge$, or more specifically, just $\neg, \wedge$ or $\neg, \vee$ (by **Corollary 5.3.4**), or even simply the connective $\downarrow$ (as we derived in the homework), and hence we don't need to come up with a name for all the connectives, especially the n -ary connectives

that have a ton of inputs and a ton of possibilities.

Note that we have 2 possible logical constants, i.e. 0-ary connectives $\top$, $\perp$

We have 4 possible unary connectives $\top$, $\perp$, $\neg$, I (2 are "effectively 0-ary"), where $I: \{T, F\} \rightarrow \{T, F\}$ is just the identity

We have 16 possible binary connectives (see page 51)

%PASTE IMAGE ON PAGE 51

The number of possible n -ary connectives as n gets larger explodes kinda quickly (there are 256 ternary connectives!). For each $n \in \mathbb{N}$, there a number of distinct n -ary connectives equal to the number of possible n -place Boolean Functions, which recall from a discussion forum reply (which I'll include in an edit of these notes) is equal to 2^{2^n} , the exponent 2^n coming from the number of possible inputs in the domain $\{T, F\}^n$ and the base 2 being the number of possible outputs in $\{T, F\}$.

Moral of the story, it's a good thing we have Boolean completeness so we don't have to make a symbol for every possible connective!